QF3101 Investment Instruments and Risk Management Computing Assignment

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Part 1: Three-Month SOFR Futures and Deferred FRA

Introduction

This part of the assignment focuses on modelling the current interest rate market environment by utilising SOFR futures and using it to construct forward SOFR term structures. The forward SOFR term structure is then used to price various deferred forward rate agreements. This part is executed in three consecutive stages with Microsoft Excel, utilising Visual Basic for Applications (VBA) for essential functional computation.

Stage 1: Data Collection and Preparation

Quotations for 3-Month SOFR Futures are available on the Chicago Mercantile Exchange (CME) with delivery months of March, June, September, and December from the period of December 2025 to March 2035. All data was collected on **15 October 2025**

Below are screenshots of the data collected:

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
AUG 2025 SR3Q5	OPT		95.79	-0.0075 (-0.01%)	95.7975	95.795	95.795	95.7875	625	09:16:50 CT 15 Oct 2025
SEP 2025 SR3U5	OPT		95.94	-0.0125 (-0.01%)	95.9525	95.95	95.9525	95.94	48,639	09:33:37 CT 15 Oct 2025
OCT 2025 SR3V5	OPT		96.1075	-0.0075 (-0.01%)	96.115	96.1125	96.115	96.105	942	08:36:59 CT 15 Oct 2025
NOV 2025 SR3X5	OPT		-	-	96.265	-	-	-	0	08:36:59 CT 15 Oct 2025
DEC 2025 SR3Z5	OPT		96.37	-0.005 (-0.01%)	96.375	96.37	96.38	96.36	146,632	09:33:32 CT 15 Oct 2025
JAN 2026 SR3F6	OPT		96.47	UNCH (UNCH)	96.47	96.47	96.47	96.47	20	08:37:00 CT 15 Oct 2025
FEB 2026 SR3G6	OPT		-	-	96.545	-	-	-	0	08:37:02 CT 15 Oct 2025
MAR 2026 SR3H6	OPT		96.605	-0.005 (-0.01%)	96.61	96.605	96.62	96.595	141,824	09:33:50 CT 15 Oct 2025
APR 2026 SR3J6	OPT		-	-	96.675	-	-	-	0	17:45:35 CT 14 Oct 2025
JUN 2026 SR3M6	OPT		96.84	UNCH (UNCH)	96.84	96.835	96.855	96.825	133,628	09:33:46 CT 15 Oct 2025
SEP 2026 SR3U6	OPT		96.98	-0.01 (-0.01%)	96.99	96.985	97.005	96.97	124,684	09:33:46 CT 15 Oct 2025

Fig 1: Aug 2025 – Sep 2026

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
DEC 2026 SR3Z6	OPT		97.04	-0.01 (-0.01%)	97.05	97.04	97.06	97.03	124,814	09:33:50 CT 15 Oct 2025
MAR 2027 SR3H7	OPT		97.04	-0.01 (-0.01%)	97.05	97.045	97.065	97.03	101,984	09:33:50 CT 15 Oct 2025
JUN 2027 SR3M7	OPT		97.02	-0.005 (-0.01%)	97.025	97.015	97.04	97.01	78,593	09:33:50 CT 15 Oct 2025
SEP 2027 SR3U7	OPT		96.98	-0.005 (-0.01%)	96.985	96.975	97.00	96.97	80,472	09:33:50 CT 15 Oct 2025
DEC 2027 SR3Z7	OPT		96.93	-0.005 (-0.01%)	96.935	96.925	96.945	96.915	63,755	09:33:46 CT 15 Oct 2025
MAR 2028 SR3H8	OPT		96.88	UNCH (UNCH)	96.88	96.87	96.895	96.86	43,792	09:33:28 CT 15 Oct 2025
JUN 2028 SR3M8	OPT		96.825	UNCH (UNCH)	96.825	96.815	96.84	96.805	44,300	09:33:38 CT 15 Oct 2025
SEP 2028 SR3U8	OPT		96.78	UNCH (UNCH)	96.78	96.765	96.795	96.76	40,148	09:32:51 CT 15 Oct 2025
DEC 2028 SR3Z8	OPT		96.735	UNCH (UNCH)	96.735	96.715	96.745	96.71	24,030	09:32:11 CT 15 Oct 2025
MAR 2029 SR3H9	OPT		96.69	UNCH (UNCH)	96.69	96.67	96.705	96.665	21,531	09:32:51 CT 15 Oct 2025
JUN 2029 SR3M9	OPT		96.65	+0.005 (+0.01%)	96.645	96.63	96.665	96.62	15,929	09:30:56 CT 15 Oct 2025
SEP 2029 SR3U9	OPT		96.605	+0.005 (+0.01%)	96.60	96.585	96.62	96.58	15,447	09:30:56 CT 15 Oct 2025
DEC 2029 SR3Z9	OPT		96.56	UNCH (UNCH)	96.56	96.545	96.575	96.53	15,384	09:31:38 CT 15 Oct 2025

Fig 2: Dec 2026 – Dec 2029

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
MAR 2030 SR3H0	OPT		96.51	UNCH (UNCH)	96.51	96.495	96.525	96.485	18,696	09:32:34 CT 15 Oct 2025
JUN 2030 SR3M0	OPT		96.465	+0.005 (+0.01%)	96.46	96.44	96.475	96.435	13,409	09:31:15 CT 15 Oct 2025
SEP 2030 SR3U0	OPT		96.415	+0.005 (+0.01%)	96.41	96.395	96.43	96.385	12,298	09:31:15 CT 15 Oct 2025
DEC 2030 SR3Z0	OPT		96.375	+0.015 (+0.02%)	96.36	96.365	96.375	96.365	269	09:30:11 CT 15 Oct 2025
MAR 2031 SR3H1	OPT		-	-	96.31	-	-	-	0	08:36:59 CT 15 Oct 2025
JUN 2031 SR3M1	OPT		-	-	96.26	-	-	-	0	08:36:59 CT 15 Oct 2025
SEP 2031 SR3U1	OPT		-	-	96.215	-	-	-	0	08:36:59 CT 15 Oct 2025
DEC 2031 SR3Z1	OPT		-	-	96.165	-	-	-	0	08:36:59 CT 15 Oct 2025
MAR 2032 SR3H2	OPT		-	-	96.12	-	-	-	0	08:36:59 CT 15 Oct 2025
JUN 2032 SR3M2	OPT		-	-	96.075	-	-	-	0	08:36:57 CT 15 Oct 2025
SEP 2032 SR3U2	OPT		-	-	96.035	-	-	-	0	08:37:03 CT 15 Oct 2025
DEC 2032 SR3Z2	OPT		-	-	95.995	-	-	-	0	08:37:01 CT 15 Oct 2025
MAR 2033 SR3H3	OPT		-	-	95.955	-	-	-	0	08:37:03 CT 15 Oct 2025

Fig 3: Mar 2030 – Mar 2033

MONTH	OPTIONS	CHART	LAST	CHANGE	PRIOR SETTLE	OPEN	HIGH	LOW	VOLUME	UPDATED
JUN 2033 SR3M3	OPT		-	-	95.915	-	-	-	0	08:37:06 CT 15 Oct 2025
SEP 2033 SR3U3	OPT		-	-	95.885	-	-	-	0	17:46:20 CT 14 Oct 2025
DEC 2033 SR3Z3	OPT		-	-	95.84	-	-	-	0	08:37:06 CT 15 Oct 2025
MAR 2034 SR3H4	OPT		-	-	95.83	-	-	-	0	17:45:46 CT 14 Oct 2025
JUN 2034 SR3M4	OPT		-	-	95.805	-	-	-	0	08:36:59 CT 15 Oct 2025
SEP 2034 SR3U4	OPT		-	-	95.79	-	-	-	0	17:46:01 CT 14 Oct 2025
DEC 2034 SR3Z4	OPT		-	-	95.745	-	-	-	0	17:45:44 CT 14 Oct 2025
MAR 2035 SR3H5	OPT		-	-	95.735	-	-	-	0	17:46:11 CT 14 Oct 2025

Fig 4: Jun 2033 – Mar 2035

Data Cleaning

From the data collected, I selected the '*Prior Settle*' prices from September 2025 to December 2034.

1. **Calculating of Implied 3-Months SOFR (%)**
   I used the formula $(100 - Futures\ Price)$ to calculate the Implied 3-months SOFR

2. **Calculating of Days in Contract Term**
   Using the settlement date and Excel's built in Date Functions, I was able to calculate the number of days within each contract.

This is the final structure of my data table:

Starting Day:	15-Oct-25			
Period	3-Month Futures Price	Implied 3-months SOFR (%)	Settlement Date	Number of Days
Sept-25	95.9525	4.0475	17-Dec-25	91
Dec-25	96.375	3.625	18-Mar-26	91
Mar-26	96.61	3.39	17-Jun-26	91
Jun-26	96.84	3.16	16-Sept-26	91
Sept-26	96.99	3.01	16-Dec-26	91
Dec-26	97.05	2.95	17-Mar-27	91
Mar-27	97.05	2.95	16-Jun-27	91
Jun-27	97.025	2.975	15-Sept-27	91
Sept-27	96.985	3.015	15-Dec-27	91
Dec-27	96.935	3.065	15-Mar-28	91
Mar-28	96.88	3.12	21-Jun-28	98
Jun-28	96.825	3.175	20-Sept-28	91
Sept-28	96.78	3.22	20-Dec-28	91
Dec-28	96.735	3.265	21-Mar-29	91
Mar-29	96.69	3.31	20-Jun-29	91
Jun-29	96.645	3.355	19-Sept-29	91
Sept-29	96.6	3.4	19-Dec-29	91
Dec-29	96.56	3.44	20-Mar-30	91
Mar-30	96.51	3.49	20-Jun-30	92
Jun-30	96.46	3.54	18-Sept-30	90
Sept-30	96.41	3.59	18-Dec-30	91
Dec-30	96.36	3.64	19-Mar-31	91
Mar-31	96.31	3.69	18-Jun-31	91
Jun-31	96.26	3.74	17-Sept-31	91
Sept-31	96.215	3.785	17-Dec-31	91
Dec-31	96.165	3.835	17-Mar-32	91
Mar-32	96.12	3.88	16-Jun-32	91
Jun-32	96.075	3.925	15-Sept-32	91
Sept-32	96.035	3.965	15-Dec-32	91
Dec-32	95.995	4.005	16-Mar-33	91
Mar-33	95.995	4.005	15-Jun-33	91
Jun-33	95.915	4.085	21-Sept-33	98
Sept-33	95.885	4.115	21-Dec-33	91
Dec-33	95.84	4.16	15-Mar-34	84
Mar-34	95.83	4.17	21-Jun-34	98
Jun-34	95.805	4.195	20-Sept-34	91
Sept-34	95.79	4.21	20-Dec-34	91
Dec-34	95.745	4.255	21-Mar-35	91

Fig 5: Final Data Table

It consists of 38 rows, with 5 variables each.

Stage 2: Implied Forward SOFR Term Structure

In this section, we calculate the implied Forward SOFR term structure for reference dates (17 Dec 2025 to 18 Dec 2030) up till 21 March 2035, I made use of a VBA-Cell Array function and the following formula:

1. **Accumulation Factor for Period i**

$$A_i = A_{i-1} \times \left[1 + \frac{d_i}{360} * \frac{r_i}{360}\right]$$

2. **Term Rate**

$$TermRate = (A_n - 1) * \frac{360}{\sum d_i}$$

Where:

- A_i is the accumulation factor for period i ($A_0 = 1$)
- r_i is the implied 3-month SOFR rate previously collected
- d_i is the number of days within the ith period

Computing the implied Forward SOFR term structures for the reference dates gives:

Reference Date		17-Dec-25	18-Mar-26	17-Jun-26	16-Sept-26	16-Dec-26	17-Mar-27	16-Jun-27	15-Sept-27	15-Dec-27	15-Mar-28	21-Jun-28	20-Sept-28	20-Dec-28	21-Mar-29	20-Jun-29	19-Sept-29	19-Dec-29	20-Mar-30	20-Jun-30	18-Sept-30	18-Dec-30
	Settlement Date	Implied Forward SOFR Term Structure (%)																				
	17-Dec-25	4.0475																				
	18-Mar-26	3.854794015	3.625																			
	17-Jun-26	3.721884258	3.523031615	3.39																		
	16-Sept-26	3.603710382	3.420781872	3.288539283	3.16																	
	16-Dec-26	3.50690369	3.337606953	3.212373667	3.097021606	3.01																
	17-Mar-27	3.43587873	3.279996242	3.164746119	3.063410616	2.991222701	2.95															
	16-Jun-27	3.388428475	3.24537916	3.140676364	3.052190724	2.992352055	2.960998993	2.95														
	15-Sept-27	3.359046186	3.227677739	3.132745572	3.055114898	3.004891219	2.980510744	2.973592205	2.975													
	15-Dec-27	3.343574504	3.222612685	3.136389436	3.067832243	3.025233797	3.006169472	3.00250313	3.006336609	3.015												
	15-Mar-28	3.339031381	3.227293622	3.148727918	3.0878006	3.051393513	3.03656815	3.03557408	3.04141908	3.065	3.065											
	21-Jun-28	3.343338995	3.240308244	3.168889795	3.11482281	3.083806317	3.072573643	3.073795811	3.081184599	3.12	3.106052477	3.12										
	20-Sept-28	3.354010906	3.258070346	3.192404853	3.143697032	3.116782404	3.10822197	3.111059499	3.119521243	3.175	3.145286976	3.159465272	3.175									
	20-Dec-28	3.368975168	3.279241076	3.218534669	3.174317768	3.150727835	3.144228073	3.148197411	3.157268598	3.22	3.182934237	3.196497555	3.210421368	3.22								
	21-Mar-29	3.387418556	3.30314665	3.246745809	3.206337499	3.185490883	3.18062449	3.185424465	3.19486702	3.265	3.22019381	3.233210469	3.2462784	3.255787643	3.265							
	20-Jun-29	3.408746145	3.329307443	3.276670887	3.239526901	3.220977957	3.217434493	3.222864891	3.232543268	3.31	3.257481538	3.27005933	3.282579873	3.292019092	3.301159037	3.31						
	19-Sept-29	3.43251348	3.357373661	3.308049652	3.273729725	3.257129609	3.254676354	3.260597319	3.270428265	3.355	3.294983394	3.307205727	3.319334744	3.328703275	3.337770105	3.346535549	3.355					
	19-Dec-29	3.458382574	3.387085028	3.340693529	3.308836909	3.293907553	3.292365493	3.298675449	3.308604748	3.4	3.332797895	3.344725179	3.356552152	3.365849304	3.374842289	3.383531442	3.391917181	3.4				
	20-Mar-30	3.485774635	3.417911538	3.374112785	3.344400318	3.330893358	3.330094865	3.336686205	3.34663967	3.44	3.370396056	3.382004447	3.393490775	3.402597695	3.411350734	3.419714846	3.427607857	3.434782444	3.44			
	20-Jun-30	3.51544103	3.450720914	3.40928203	3.381503901	3.369271792	3.369139452	3.375989812	3.38598171	3.49	3.409430128	3.420826607	3.432116834	3.4417244	3.44988696	3.45824172	3.46619912	3.473671071	3.480393273	3.49		
	18-Sept-30	3.546229891	3.484305081	3.444955164	3.418876338	3.407758278	3.408208471	3.415291164	3.425329759	3.54	3.44856755	3.459826023	3.471008533	3.480079545	3.488836574	3.497278947	3.505405205	3.513212029	3.520690989	3.530338231	3.54	
	18-Dec-30	3.578960788	3.519613782	3.482181597	3.45765181	3.447543658	3.448520839	3.455816167	3.465903414	3.59	3.488982021	3.500140353	3.511256274	3.520364711	3.529184697	3.537727813	3.546013153	3.554074762	3.561980211	3.57158338	3.581111638	3.59
	19-Mar-31	3.613164425	3.55617341	3.520486365	3.497356941	3.48815662	3.48960573	3.497092101	3.507225635	3.64	3.530178068	3.541259597	3.552330667	3.56148038	3.570361882	3.578993984	3.587406069	3.595646832	3.603803509	3.613334436	3.622739683	3.631515994
	18-Jun-31	3.648735346	3.593902966	3.559808343	3.537948956	3.529567396	3.531440068	3.539097507	3.549273202	3.69	3.572120146	3.583139054	3.594176279	3.603363717	3.61229799	3.621001823	3.629509149	3.637873052	3.646181656	3.655630251	3.664921141	3.67359264
	17-Sept-31	3.685587941	3.632737888	3.600099968	3.579395776	3.571754837	3.574008423	3.581818615	3.592031424	3.74	3.614785586	3.625750388	3.636758666	3.645977057	3.654952958	3.663710935	3.672286334	3.680731963	3.689130502	3.698491811	3.70767293	3.716241782
	17-Dec-31	3.723408432	3.672375027	3.641064166	3.621403996	3.61442421	3.617009213	3.624942412	3.635172373	3.785	3.657806831	3.668701874	3.679657156	3.688871184	3.697845375	3.706603634	3.715179299	3.723620335	3.731998424	3.741180838	3.75013497	3.758437953
	17-Mar-32	3.762398412	3.713040542	3.682949594	3.664244555	3.657866674	3.660751638	3.66879506	3.679039441	3.835	3.701565467	3.712398129	3.723305997	3.732514689	3.741485284	3.750240924	3.758813046	3.767245493	3.775601351	3.784641956	3.793429939	3.801560296
	16-Jun-32	3.802279607	3.754463284	3.725488554	3.707651103	3.701813388	3.704960281	3.713091249	3.723334924	3.88	3.745732132	3.756490889	3.767334395	3.776514206	3.78545373	3.794174572	3.802705428	3.811085272	3.819368449	3.828231896	3.836814163	3.844724418
	15-Sept-32	3.843031284	3.796635816	3.768684531	3.751637045	3.746285335	3.749661318	3.757860752	3.768091463	3.925	3.790344542	3.801021047	3.811787867	3.820921559	3.829810467	3.838474396	3.846939147	3.855238972	3.863420381	3.872092425	3.880456782	3.888136214
	15-Dec-32	3.88441862	3.83932937	3.812311988	3.795978471	3.791057451	3.794625705	3.802868374	3.81306622	3.965	3.835140926	3.845715998	3.856381371	3.865439789	3.87424557	3.882816303	3.891174512	3.899349514	3.907380211	3.915819099	3.923919301	3.931316088
	16-Mar-33	3.926445126	3.882558704	3.856395129	3.84070828	3.836169667	3.839898786	3.848163894	3.858313103	4.005	3.880183831	3.890643556	3.901189653	3.910151839	3.91885173	3.927304654	3.935529938	3.9435523	3.951403894	3.95959005	3.967407468	3.974504174
	15-Jun-33	3.967444135	3.924628845	3.899204373	3.884057285	3.879807001	3.883613526	3.89182241	3.901843802	4.005	3.923340212	3.93358836	3.943900106	3.952645805	3.961107138	3.969295252	3.977223993	3.984910781	3.992377853	4.000085492	4.007357937	4.01385551
	21-Sept-33	4.014022821	3.972306857	3.947651587	3.933076291	3.929150108	3.933074593	3.941272886	3.951217281	4.085	3.972458666	3.982564867	3.992725972	4.00134207	4.009663087	4.017698369	4.025459371	4.032960281	4.040218904	4.047660746	4.054647325	4.060854585
	21-Dec-33	4.057562248	4.016778799	3.992767906	3.978663825	3.974993279	3.978994376	3.987157342	3.997007892	4.115	4.01797007	4.027921366	4.037914856	4.046381054	4.054540092	4.062399501	4.069968402	4.077257942	4.0842819	4.091437561	4.098115251	4.104005202
	15-Mar-34	4.098655616	4.058677123	4.035221514	4.021522496	4.018071008	4.022137239	4.030269453	4.040039419	4.16	4.060763639	4.07058382	4.080436256	4.088777174	4.096801824	4.104516542	4.111928922	4.11904814	4.125885396	4.132816102	4.139255638	4.144906814
	21-Jun-34	4.145942472	4.106807309	4.083918045	4.070617429	4.067357973	4.071445371	4.079491295	4.089117516	4.17	4.109459953	4.119069761	4.128690245	4.136816001	4.144609433	4.152074847	4.159217332	4.166042948	4.172558966	4.17911682	4.185154291	4.190389956
	20-Sept-34	4.190047054	4.151625968	4.129208457	4.116232089	4.113116702	4.117199219	4.125146369	4.134623165	4.195	4.154582168	4.163981637	4.173369387	4.181277552	4.188838098	4.196053576	4.202926958	4.20946171	4.215661898	4.221859441	4.227510232	4.232347817
	20-Dec-34	4.233974186	4.196199904	4.174199478	4.161499689	4.15849024	4.162537705	4.170360522	4.179664139	4.21	4.199192993	4.208357307	4.217483721	4.225145432	4.232442952	4.239377077	4.245948679	4.252158708	4.25800818	4.263811432	4.269039658	4.273442069
	21-Mar-35	4.278871028	4.241700798	4.220088984	4.207645999	4.204733839	4.208747174	4.216452958	4.225594511	4.255	4.244723251	4.253670089	4.262555932	4.269991966	4.27705	4.283729573	4.290030078	4.295950711	4.301490401	4.306948574	4.31181147	4.315843027

Fig 6: Term Structure Table

Analysis of Forward Term Structure

Using Excel's built-in visualisation tools, I created seven graphs with consecutive reference dates of:

- 17 December 2025
- 18 March 2026
- 17 June 2026
- 16 September 2026
- 16 December 2026
- 17 March 2027
- 16 June 2027

The graphs plot the Implied Forward SOFR Term Structures for each reference date against the different settlement dates (each representing one quarter)

I obtained the following graphs:

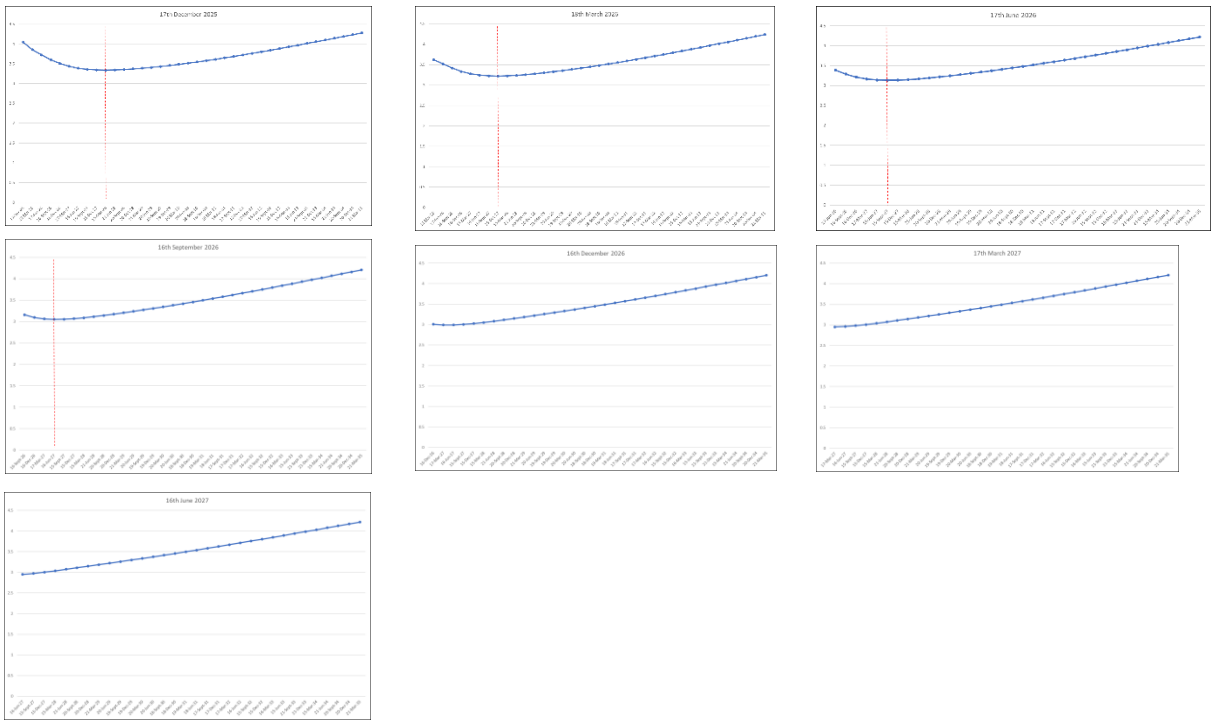

Fig 7: Implied Forward SOFR Term Structures Graphs

Reference Date	Analysis
17th December 2025	<u>From the Reference date to March 2028, the curve exhibits a steep decreasing non-linear trend</u> The curve is initially decreasing from the reference date, 17th December 2025, out to approximately 2.25 years later, March 2028. This decrease is a reliable market signal predicting monetary policy easing (rate cuts) within the next two years. This prices in the expectation of a subsequent recession or significant economic slowdown, which will encourage the central bank to cut the SOFR rate to stimulate aggregate demand. The steepness of this initial decline reflects the market's sentiments regarding the velocity and magnitude of the rate cuts. <u>Turning Point at March 2028, at this point, rates are also at the lowest in the entire period</u> The point of minimum yield marks the turning point of the forecasted yield curve. The market forecasts the end of economic weakness and of the central bank's rate-cutting cycle. After this point, rates are expected to stabilise and then rise. <u>From March 2028 onwards, the curve exhibits a non-linear increasing trend</u> This segment is primarily driven by the Term Premium (compensation for duration risk) and the expectation that the economy will eventually return to a state of long-term stability and normalised growth. Lenders will demand for higher returns for tying up capital for longer durations to compensate for inflation uncertainty and illiquidity risk over extended periods.
18th March 2026	<u>From the reference date to December 2027, the curve exhibits a decreasing non-linear trend. This curve is not as steep as the one for 17th December 2025</u> The curve is initially decreasing for a period of approximately 1.75 years. Similar to the previous graph, this decrease is a reliable market signal predicting monetary policy easing (rate cuts) within the next two years. The high current short-term rates are a result of the central bank's restrictive policy to curb inflation. As this graph is a for a period closer to the end of the rate cuts, the steepness of the curve will not be as great as the previous curve. <u>Turning point at Dec 2027, earlier than the turning point of the previous turning point</u> The earlier turning point indicates increased market confidence that the policy easing cycle will begin sooner and that the period of economic weakness will bottom out earlier than previously expected. <u>From December 2027 onwards, the curve exhibits a non-linear increasing trend</u> Similar to the previous graph, the increasing slope reflects the required Term Premium and the expectation that SOFR rates will rise as the economy recovers.
17th June 2026	<u>From the reference date to September 2027, the curve exhibits a decreasing non-linear trend. This curve is not as steep as the one for the previous 2 curves</u> The curve is initially decreasing for a period of approximately 1.5 years. Similar to the previous graph, this decrease is a reliable market signal predicting monetary policy easing (rate cuts) within the next two years. The high current short-term rates are a result of the central bank's restrictive policy to curb inflation. As this graph is a for a period closer to the end of the rate cuts, the steepness of the curve will not be as great as the previous two curve. <u>Turning point at September 2027, earlier than the turning point of the previous turning point</u> Similarly, the earlier turning point indicates increased market confidence that the policy easing cycle will begin sooner and that the period of economic weakness will bottom out earlier than previously expected. <u>From September 2027 onwards, the curve exhibits a non-linear increasing trend</u> Similar to the previous graph, the increasing slope reflects the required Term Premium and the expectation that SOFR rates will rise as the economy recovers. However, The gentler curve indicates that the market anticipates a slower, more measured increase in SOFR rates.

16th September 2016	From the reference date to September 2027, the curve exhibits a decreasing non-linear trend. However, this curve is short-lived and is has become much less pronounced. The decreasing trend within the period is a signal that the market expects the start of the rate-cutting cycle to occur in the near future. The curve's flatness signifies that the current high rates are expected to drop immediately, narrowing the difference between short-term and medium-term yields. This suggests a consensus of reduced volatility—the market is no longer debating if rates will drop, but is merely waiting for the action to begin. Turning point at June 2027, earlier than the turning point of the previous turning point Similarly, the earlier turning point indicates increased market confidence that the policy easing cycle will begin sooner and that the period of economic weakness will bottom out earlier than previously expected. From June 2027 onwards, the curve exhibits a non-linear increasing trend Similar to the previous graph, the increasing slope reflects the required Term Premium and the expectation that SOFR rates will rise as the economy recovers. The long-term trend now visually dominates the front-end inversion. This reflects the market views that the bulk of the required rate cuts has been priced into the yield curve, and the dominant remaining factor is the long-term cost of borrowing.
16th December 2026	From the reference date to June 2027, the curve is nearly flat The flatness of the curve indicates the market expects interest rates over the next 12 months to be stable and predictable. The expected rate cuts are no longer a significant event because the market has already fully priced in the expected policy easing. The current short-term rate is seen as the stabilised rate that will hold until the recovery begins, eliminating the steepness seen in prior periods. From June 2027 onwards, the curve exhibits a non-linear increasing trend The positive slope after the first year signifies that the market is now pricing the economic recovery and subsequent policy normalisation. The rate of increase is determined by the forecast for a strong rebound and the increasing Term Premium. The market is expecting an increase in premium for duration risk, reflecting a growing belief that long-term inflation risk is high.
17th March 2027	From the reference date onwards, the curve exhibits an increasing trend that is almost linear. There is almost no period of flatness or a turning point as seen in previous graphs The immediate and sustained upward slope confirms that the market views the current short-term rate as being at or near the definitive trough. As the period is at a later date as compared to the previous graphs, the lack of a decrease or turning point indicates that the market believes the central bank's easing policies has successfully concluded. The linearity of the curve is a strong economic indicator that the curve is now primarily driven by the Pure Term Premium. Every additional period of duration risk (e.g., every quarter) requires an equal, constant amount of compensation from the market. Furthermore, The absence of any initial humps, dips, or exaggerated steepness shows that short-term funding risk and policy uncertainty are very low. The market is highly confident that funding conditions will remain stable and orderly for the foreseeable future.
16th June 2027	From the reference date onwards, the curve exhibits an increasing trend that is near-linear. There is no period of flatness or a turning point as seen in previous graphs The complete absence of flatness or an initial decrease in rates shows short-term volatility and policy risk no longer play a role in affecting market sentiments. The market is entirely confident that the central bank is done cutting rates and is either on hold or already beginning the slow process of rate hikes. There is no uncertainty priced into the front end. The immediate and consistent upward slope confirms that the market views the current short-term rate as being firmly above the policy trough. The entire forecast for policy intervention (rate cuts, stability periods, and turning points) is now complete, and the sole remaining economic dynamic is the rise toward equilibrium.

Overall Insights from Analysis of Implied Forward SOFR Term Structures

1. Initial Decreasing Trend:
The fact that the short-term rates are higher than medium-term rates suggests that the market expects monetary policy easing to occur within the next one to two years. The is market speculation that the current short-term rates are high in order to slow economic growth and reduce inflation. The market anticipates a subsequent recession or significant economic slowdown, forcing the central bank to cut the SOFR rate to stimulate the economy.

2. Turning Point:
The turning point is generally around mid-late 2027 and this represents the market's expectation of when the central bank will stop cutting rates and when the cycle of economic weakness will bottom out and stabilise.

3. Long-Term Increasing Trend:
The consistent upward slope in the long-term segment of all curves (March 2028 onwards) is characteristic of a normal yield curve. This reflects the Term Premium and the expectation of long-term economic growth and normalised inflation. Lenders require higher returns for tying up capital for longer periods in order to compensate for risk and inflation uncertainty over longer periods.

The comparison of the term structures across the seven reference dates reveals a crucial market transition: the movement from a decreasing curve (recession-predicting) to a normal increasing curve (growth-predicting).

Stability of Long-Term Rates:
The rates for the longest maturities (Mar 2032 - Mar 2035) are relatively stable and consistent across all seven reference dates, with rates settling within the same range of values for all the settlement dates within this period. This indicates that long-term inflation and economic growth expectations remain anchored and do not fluctuate significantly with short-term policy changes or market speculations.

Dynamic Changes in the Short-to-Medium Term:
The most dramatic change is the decreasing, flattening and eventual normalization of the front end (until 2028). As the reference date moves from 17 Dec 2025 to 16 Jun 2027, the initial decrease becomes shallower and eventually disappears. This suggests the market is expecting the anticipated rate cuts to occur sooner, or is growing more confident that the period of high interest rates will be short-lived. By March 2027, the purely upward trending graph indicates that the market may believe the central bank's restrictive policy has either done its job or that inflation risks have subsided, leading to a stabilisation of expected rates over the remaining period.

Further Comments and Insights

The transition of the term structure from a decreasing to increasing trending curve provides a clear narrative of the market's changing risk assessment:

1. Risk Premium
The initial decrease incorporates a high degree of recession risk premium. As the curve flattens toward 2027, this near-term risk premium diminishes, suggesting less concern about an imminent, sharp economic downturn.

2. Monetary Policy Signal
The term structure is heavily influenced by the expectations of the SOFR rates. The fact that all long-term zero rates are positive and relatively stable implies the market believes the long-run interest rate is healthy and positive, reflecting a return to moderate long-term growth.

3. Trading Strategy
For investors, the initial steep decline in the forward rates could suggest a favourable strategy of buying long-term fixed-income instruments or receiving fixed rates in FRA contracts. This would be beneficial as short-term rates fall. As the curve normalises, this strategy becomes less profitable.

Stage 3: Pricing Deferred Forward Rate Agreement

In this section we calculate the fixed rate (per annum, actual/360) of a deferred I × J FRA with a starting date of 17 Dec 2025, using the Implied Forward SOFR Term Structure for 17 December 2025 and an excel VBA user-defined function,

I made use of the following formulas:

1. **Calculate Days in Period I / J:**

$$D_I = \sum_{i=1}^{\frac{I}{3}} d_i$$

$$D_J = \sum_{i=1}^{\frac{J}{3}} d_i$$

2. **Adjusting for actual / 360 day count convention:**

$$T_I = \frac{D_I}{360}$$

$$T_J = \frac{D_J}{360}$$

3. **Discount Factors**

$$DF_I = \left(1 + \frac{r_I}{100} * T_I\right)^{-1}$$

$$DF_J = \left(1 + \frac{r_J}{100} * T_J\right)^{-1}$$

4. **Forward Rate Pricing**

$$F(I,J) = \left(\frac{DF_I}{DF_J} - 1\right) * \left(\frac{T_J - T_I}{360}\right)^{-1} * 100$$

Where:

- r_i is the implied 3-month SOFR rate previously collected
- d_i is the number of days within the ith period

Computing the FRA Rates for $I \in \{3, 6, 9, \dots, 90\}$ and $J = I + 3n$, for $n \in \{1,2,3,4\}$ gives:

I	J	FRA
3	6	3.554341
	9	3.60371
	12	3.506904
	15	3.435879
6	9	3.305172
	12	3.506904
	15	3.435879
	18	3.388428
9	12	3.130921
	15	3.435879
	18	3.388428
	21	3.359046
12	15	3.043848
	18	3.388428
	21	3.359046
	24	3.343575
15	18	3.02003
	21	3.359046
	24	3.343575
	27	3.339031
18	21	3.027182
	24	3.343575
	27	3.339031
	30	3.343339
21	24	3.053768
	27	3.339031
	30	3.343339
	33	3.354011
24	27	3.09352
	30	3.343339
	33	3.354011
	36	3.368975
27	30	3.140757
	33	3.354011
	36	3.368975
	39	3.387419
30	33	3.189892
	36	3.368975
	39	3.387419
	42	3.408746
33	36	3.231277
	39	3.387419
	42	3.408746
	45	3.432513
36	39	3.273488
	42	3.408746
	45	3.432513
	48	3.458383
39	42	3.316308
	45	3.432513
	48	3.458383
	51	3.485775
42	45	3.359585
	48	3.458383
	51	3.485775
	54	3.515441
45	48	3.403211
	51	3.485775
	54	3.515441
	57	3.54623
48	51	3.44235
	54	3.515441
	57	3.54623
	60	3.578961
51	54	3.491223
	57	3.54623
	60	3.578961
	63	3.613164
54	57	3.540281
	60	3.578961
	63	3.613164
	66	3.648735
57	60	3.58953
	63	3.613164
	66	3.648735
	69	3.685588
60	63	3.638922
	66	3.648735
	69	3.685588
	72	3.723408
63	66	3.688433
	69	3.685588
	72	3.723408
	75	3.762398
66	69	3.738041
	72	3.723408
	75	3.762398
	78	3.80228
69	72	3.782895
	75	3.762398
	78	3.80228
	81	3.843031
72	75	3.83264
	78	3.80228
	81	3.843031
	84	3.884419
75	78	3.877594
	81	3.843031
	84	3.884419
	87	3.926445
78	81	3.922579
	84	3.884419
	87	3.926445
	90	3.967444
81	84	3.962728
	87	3.926445
	90	3.967444
	93	4.014023
84	87	4.002882
	90	3.967444
	93	4.014023
	96	4.057562
87	90	4.004031
	93	4.014023
	96	4.057562
	99	4.098656
90	93	4.083245
	96	4.057562
	99	4.098656
	102	4.145942

Fig 8: Table for FRA Rates (with respective I and J values)

Analysis of FRA Rates

Using Excel's built-in visualisation tools, I created graphs for the FRA rates for the different FRA contracts.

- Fixed n, Vary I (30 graphs, one for each value of $I \in \{3, 6, 9, \dots, 90\}$)
    - Each graph consists of Fixed Rates Plotted against n

- Fixed I, Vary n (4 graphs, one for each value of $n \in \{1,2,3,4\}$)
    - Each graph consists of Fixed Rates plotted against the values of I

1. Fixed n, Vary I

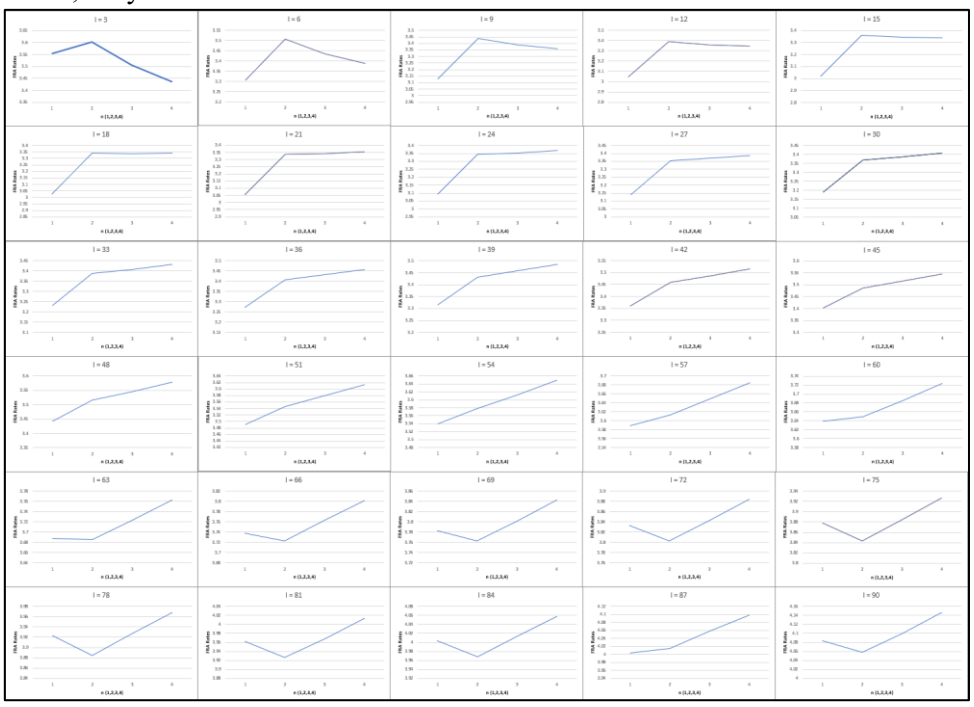

Fig 9: FRA Rates graph for varying I

2. Fixed I, Vary n

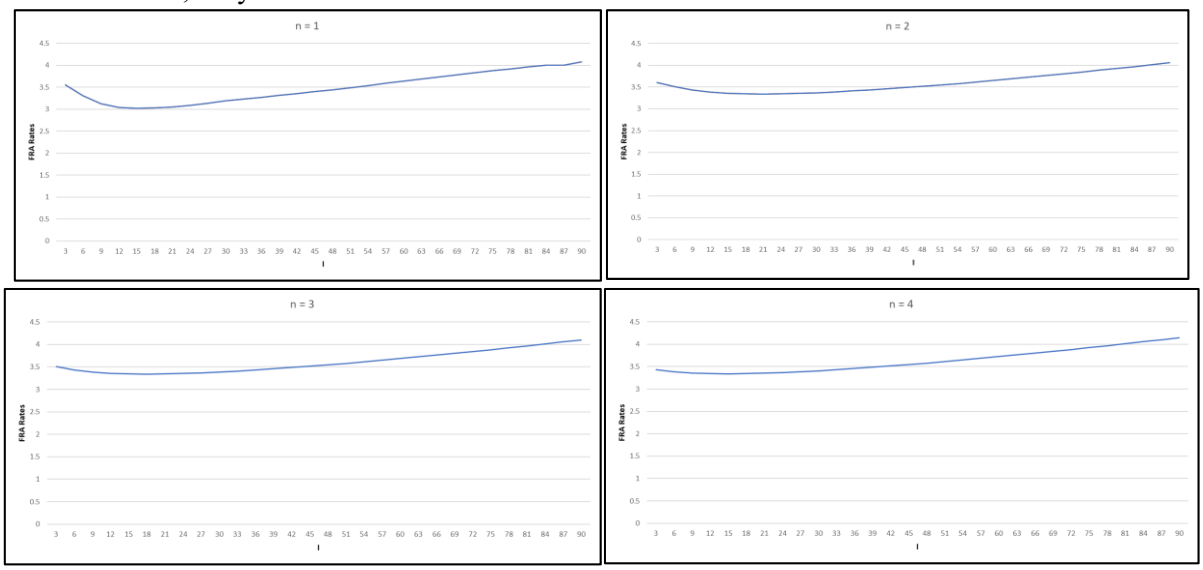

Fig 10: FRA Rates graph for varying n

Analysis for Graphs of Fixed *n*, Varying *I*

Values of I	Analysis
I = 3 to I = 15	<u>From n = 1 to n = 2 there is a steep increase in FRA rates</u> It suggests the market views the period immediately following the initial 3 months after I as having unexpectedly high-interest rates. It indicates a delay in the anticipated policy easing, or a higher volatility premium applied to the 6-month period, reflecting uncertainty about the exact timing and speed of the initial rate cuts. The market is pricing a brief spike in borrowing costs. <u>At n = 2, there is a turning point, this point is also the highest value that the FRA rate will achieve.</u> This reflects contracts that last from a 9-month to a 21-month period, roughly until September 2027. The peak at the 6-month term signifies the maximum short-term volatility priced into the contract. The market is pricing the highest FRA rate at this point, and speculates that after this point, the uncertainty is resolved and the central bank's actions become predictable. This marks a critical transition point in market sentiment. <u>At n = 2 to n =4, the FRA rates decline. For I = 3, the rate decrease is the steepest, and the decrease gradually smoothens out.</u> This steep decline is the reflective of the market forecasting for a deep and sustained rate-cutting cycle. Extending the contract to a 9 or 12-month term averages the initial high rate with the significantly lower expected rates in the distant future. The decline is steepest for I=3 because that FRA contract starts closest to the current high policy rate, meaning the 12-month average absorbs the largest possible change. The smoother decline for I=15 shows that by the time the market reaches that deferred start date, much of the rate cut has already been priced and the volatility to be averaged out is lower, signalling greater rate stability near the expected turning point.
I = 18 to I = 30	<u>From n = 1 to n = 2, the steep increase in FRA rates is still observable and is similar to the ones from the I=3 to I=15 curves</u> This curve still reflects a market speculation of high interest rates. Thus, there is a high liquidity/volatility premium applied to the 6-month term. The market maintains caution when extending contracts from 3 to 6 months, demonstrating a belief that short-term rate volatility around the low point of the cycle will remain high. <u>At n = 2, there is a turning point, however this point is no longer the highest value that the FRA rate will achieve.</u> This reflects contracts that last from a 24-month to a 36-month period. The fixed start I is now placed at or near the market-forecasted rate turning point. Therefore, extending the contract term means the contract absorbs a period dominated by the implied SOFR forward term structure positive slope (recovery) rather than its negative slope (recession). This confirms that the market views this time window as the transition from policy management to fundamental economic growth. <u>At n = 2 to n =4, there is a gentle increase in FRA rates. For I = 18, the rate increase is almost flat. The slope of this increase gradually rises as I increase from I = 18 to I = 30. The Final FRA rates are higher or equal to the peak FRA rates when n = 2</u> The increasing rate is driven by two factors: - The expectation that SOFR rates will rise as the economy recovers - The addition of the Term Premium for increased duration risk. The rate increase confirms that the Term Premium and recovery expectations now dominate the FRA price, overriding any residual short-term policy volatility.

I = 33 to I = 51	<u>From n = 1 to n = 2, the increase in FRA rates is still observable but it gradually becomes more and more gentle as I increase from I = 33 to I = 51</u> The increasing gentleness signifies that the short-term uncertainty that caused the spikes in the previous graphs are gradually easing. The market views the rate environment 33 to 51 months out as highly stable and predictable, reducing the risk premium applied to extending the contract from 3 months to 6 months. This confirms strong confidence in the stability of long-term funding conditions. <u>At n = 2, there is a turning point.</u> The turning point is merely a mechanical signal here. Since this segment (I=33 to 51 months) is entirely within the market's expected recovery and long-term stability phase. The rate cuts are over and the market expects the n = 2 rate to always be surpassed by n = 3 and n = 4 rates. The market is expecting that at these maturities, rates only go up as the term lengthens. <u>At n = 2 to n =4, there is still an increase in FRA rates. The steepness of the slope rises as I increases from I = 33 to I = 51</u> The consistent rise confirms that FRA pricing is now fully dominated by long-term risk factors. Extending the term adds the Term Premium required for long-duration exposure and aligns the FRA rate with the market's projected long-term rate. The steepening trend indicates that the the market anticipates that FRA contracts that start later will experience a faster and stronger upward pressure on rates compared to those starting earlier. This implies that economic conditions are expected to accelerate the central bank's rate hikes as time progresses through this long-term recovery phase.
I = 54	This is a unique and important point as the graph appears almost fully linear. <u>From n = 1 to n = 4, FRA rates increase steadily, there is no observable turning point at n=2 as seen in previous graphs</u> The complete absence of an observable turning point and the linear slope confirms that the FRA is now fully free from short-term monetary policy uncertainty. The market considers the entire policy cycle—rate cuts, turning points, and recovery—to be concluded and fully priced before I = 54. This indicates a high level of confidence in the long-term forecasts of rates.
I = 57 to I = 60	<u>From n = 1 to n = 2, the increase in FRA rates is still observable but it gradually becomes more and more gentle as I increase from I = 57 to I = 60. It is becoming near-flat</u> The near-flatness of the curve at this point reflects the market views that the interest rates over this time horizon will be fully stable and predictable. The short-term volatility premium that dominated earlier maturities (I<15) has essentially vanished. The cost of borrowing for 3 months versus 6 months, starting 5 years out, is practically identical, signalling a complete absence of perceived short-term funding risk. <u>At n = 2, the turning point returns, there is an observable increase in the steepness of the rate after this point</u> This marks the transition where the FRA pricing shifts from incorporating minimal, short-term duration cost to incorporating the full, accelerating cost of long-term duration risk. Although both segments (n = 1 to n = 2) and (n = 2 to n = 4) demonstrate an increasing trend, the steepness of the latter curve is much stronger, indicating a shift in the factors that affect FRA rates. <u>At n = 2 to n =4, there is still an increase in FRA rates. For the first time, the steepness of the slope here is greater than that of the slope from n=1 to n=2.</u> While the market is comfortable pricing the first six months, extending the contract from 6 months to 12 months incurs a significantly higher premium. This reflects the reality that uncertainty—particularly regarding long-term inflation and the nominal growth rate—compounds disproportionately over a long period. The market demands a higher premium for taking on this extended, deep long-term risk.

I = 63 to I = 90	<u>From n = 1 to n = 2, there is a decrease in FRA rates. This decrease grows steeper as I increass from I = 63 to I = 90, I = 87 is an outlier to this trend.</u> This return to a decreasing trend indicates the market is forecasting a possible policy-easing phase within the distant future. The expected rate for the first 3 months (n = 1)is higher than the average for the 6 months (n = 2). This suggests the market expects the highest SOFR rate to be achieved soon after the contract starts, followed by a minor expected rate cut or policy relief shortly thereafter. The increasing steepness confirms this forecast becomes more pronounced the further out the starting date is deferred. <u>At n = 2, the turning point becomes more significant. The FRA rate at this point is the lowest is will achieve. I = 87 is an outlier to this trend.</u> The return of n = 2 as the lowest rate confirms that the market is pricing a temporary, localised turning point immediately after I+6. The market expects rates to briefly fall and then immediately resume their long-term climb, minimising the duration of the easing policy. <u>At n = 2 to n =4, the increase in FRA rates is steep. The steepness of this slope increases from I = 63 to I = 90.</u> This trend is the strongest signal of duration risk cost. After the brief policy softness (n = 2), extending the term drastically increases the rate. The extreme, accelerating steepness confirms that the market demands a rapidly increasing premium for taking on uncertainty, inflation risk, and liability matching over these most distant horizons. This segment of the curve is essentially a measure of the market's maximum required risk compensation. <u>Outlier at I = 87</u> The graph at I=87 does not follow the trends of the graphs before and after it. This could be due to a data anomaly or illiquidity. At such extreme maturities (almost 8 years out), markets are highly illiquid. This outlier could be due to a data anomaly or a lack of trading volume leading to a rate quote that deviates from the trends previously seen. Economically, the pricing mechanism is being overwhelmed by low confidence and minimal volume, making the I = 87 point statistically unreliable for economic inference.

Analysis for Graphs of Fixed *I*, Varying *n*

Values of *n*	Analysis
$n = 1$	As I increases from 3 months to 15 months, there is a steep decrease in FRA rates This initial decline is reflective of the market speculating an imminent central bank rate-cutting cycle. The high current short-term rates (reflecting the implied Forward SOFR term structure curve) are expected to fall rapidly due to anticipated economic slowdown or recessionary pressure. At I = 15, there is a turning point. This corresponds to a 15 x 18 FRA Contract, this contract covers the period up till June 2027 This point represents the market's consensus forecast that signals the end of expected rate cuts, after which rates are expected to stabilise or begin to rise. A steady, strong increase as I increases past 15 months up to 90 months. The final rate is higher than the initial rate at I=3 This increasing segment reflects the market's expectation of long-term nominal economic growth and a return to a healthy, positive equilibrium interest rate. The long-term FRA rate incorporates a positive Term Premium (compensation for risk and inflation uncertainty).
$n = 2$	As I increases from 3 (3 x 9 FRA) to 21 (21 x 27 FRA), there is a gentler decreasing trend in FRA rates. This slope is less steep than the n = 1 curve This initial decline is reflective of the market speculating an imminent central bank rate-cutting cycle. However, because this is a 6-month average, the impact of any sharp, short-lived quarter-by-quarter rate changes is smoothed out leading to a gentler decreasing slope. The market sees the immediate rate cutting, but the longer average term dilutes the quarterly volatility. The turning point occurs slightly later, at I = 21. This corresponds to a 21 x 27 FRA Contract, this contract covers the period up till March 2028 This suggests that when averaging over six months, the market consensus forecasts the bottom of the full rate-cutting cycle slightly later than the n=1 curve. This point still signifies the expected end of recessionary policy, after which rates are expected to stabilise or begin to rise. A steady, strong increase as I increases past 21 months up to 90 months. The final rate is higher than the initial rate at I=3. The slope appears smoother than the n=1 curve This increasing segment reflects the market's expectation of long-term nominal economic growth and a return to a healthy, positive equilibrium interest rate. The smoother rise, compared to the n=1 curve, demonstrates the averaging effect, the 6-month FRA is less sensitive to the expected rate volatility within any single future 6-month period.
$n = 3$	As I increases from 3 (3 x 12 FRA) to 18 (18 x 27 FRA), there is a very gentle decreasing trend in FRA rates. This slope is less steep than the n=2 curve This initial decline is reflective of the market speculating an imminent central bank rate-cutting cycle. However, because this is a 9-month average, it further smoothens out the impact of any sharp, short-lived quarter-by-quarter rate changes, leading to a gentler decreasing slope. This suggests minimal concern about the pace of the expected policy easing. The turning point occurs at slightly earlier than the n = 2 curve, at I = 18. But this corresponds to a 18 x 27 FRA Contract, which also covers the period up till March 2028 This suggests that when averaging over nine months, the market consensus forecast for the bottom of the full rate-cutting cycle is similar for when averaging over six months. This point still signifies the expected end of recessionary policy, after which rates are expected to stabilise or begin to rise. A steady, strong increase as I increases past 18 months up to 90 months. The final rate is higher than the initial rate at I=3. The slope appears similar to the n=2 curve This increasing segment reflects the market's expectation of long-term nominal economic growth and a return to a healthy, positive equilibrium interest rate. This rise is consistent with the curve for when n = 1 and n = 2, indicating the stability of the long-term premiums.

$n = 4$	For n = 4: As I increases from 3 (3 x 18 FRA) to 15 (15 x 27 FRA), there is a very gentle, almost flat decreasing trend in FRA rates. This slope is the flattest of all 4 curves This initial decline is reflective of the market speculating an imminent central bank rate-cutting cycle. The near-flatness of the decline indicates that a 12-month average might be long enough to smooth out the expected volatility associated with the quarter-by-quarter rate cuts. The FRA pricing reflects expectations of medium-term stability in rates instead of a sharp decline. This suggests that the market is pricing for a high probability that there will be a smooth and orderly transition in monetary policies. The turning point occurs at slightly earlier than the n = 3 curve, at I = 15. But this corresponds to a 15 x 27 FRA Contract, which also covers the period up till March 2028 This suggests that when averaging over twelve months, the market consensus forecast for the bottom of the full rate-cutting cycle is similar for when averaging over nine months. This point still signifies the expected end of recessionary policy, after which rates are expected to stabilise or begin to rise. A steady, strong increase as I increases past 18 months up to 90 months. The final rate is higher than the initial rate at I=3. The slope appears similar to the n = 2 curve This increasing segment reflects the market's expectation of long-term nominal economic growth and a return to a healthy, positive equilibrium interest rate. This rise is consistent with the curve for when n = 1, n = 2 and n = 3, indicating the stability of the long-term premiums.

- End of Part 1 -

Part 2: Pricing an Interest Rate Swap and Tracking its Value

Introduction

This part of the assignment focuses on Historical Analysis using the former USD LIBOR in 2010. The term structure of each of these historical dates is used to obtain the swap rate of a 12-month interest rate swap. The swap rate is then used to calculate and valuate the swap at different monthly intervals.

Stage 1: Data Collection and Preparation

Historical USD LIBORs (quoted on actual/360) for maturities up to 12 months were collected for D.

Where D is the set of dates given:

- January 04 2010
- February 01 2010
- March 01 2010
- April 01 2010
- May 03 2010
- June 01 2010
- July 01 2010
- August 02 2010
- September 01 2010
- October 01 2010
- November 01 2010
- December 01 2010

These are the first business days of the month in the United States in 2010.

D	4-Jan-10	1-Feb-10	1-Mar-10	1-Apr-10	3-May-10	1-Jun-10	1-Jul-10	2-Aug-10	1-Sept-10	1-Oct-10	1-Nov-10	1-Dec-10
Maturity(mont	Term Structure	Term Structure2	Term Structure3	Term Structure4	Term Structure5	Term Structure6	Term Structure7	Term Structur	Term Structure9	Term Structu	Term Structu	Term Structure1
1 month rate	0.23344%	0.22906%	0.22813%	0.24863%	0.28469%	0.35088%	0.34719%	0.30281%	0.25781%	0.25688%	0.25375%	0.26531%
2 month rate	0.24344%	0.23906%	0.23938%	0.26694%	0.32125%	0.43438%	0.43063%	0.36094%	0.27469%	0.27359%	0.26859%	0.28250%
3 month rate	0.25438%	0.24906%	0.25169%	0.29150%	0.35313%	0.53625%	0.53331%	0.44469%	0.29563%	0.29063%	0.28594%	0.30344%
4 month rate	0.29125%	0.27625%	0.28563%	0.32850%	0.40500%	0.60775%	0.60338%	0.51213%	0.35594%	0.34469%	0.33500%	0.35031%
5 month rate	0.35025%	0.32438%	0.32669%	0.37063%	0.46375%	0.67119%	0.67163%	0.58125%	0.42819%	0.40813%	0.39625%	0.41094%
6 month rate	0.43438%	0.38375%	0.38375%	0.44156%	0.54578%	0.76113%	0.75100%	0.65994%	0.49625%	0.46313%	0.44563%	0.46656%
7 month rate	0.51250%	0.45563%	0.45481%	0.51563%	0.62531%	0.83213%	0.81819%	0.71944%	0.55350%	0.51344%	0.49350%	0.51875%
8 month rate	0.62000%	0.53438%	0.52750%	0.58906%	0.70375%	0.90363%	0.88656%	0.77375%	0.60888%	0.56375%	0.54125%	0.57125%
9 month rate	0.71813%	0.61313%	0.60138%	0.67125%	0.78688%	0.98719%	0.95375%	0.82781%	0.65813%	0.61203%	0.59313%	0.62156%
10 month rate	0.80875%	0.69125%	0.67931%	0.75294%	0.86719%	1.05863%	1.02500%	0.89656%	0.71575%	0.66531%	0.64750%	0.67313%
11 month rate	0.90063%	0.76563%	0.75938%	0.83375%	0.94906%	1.13675%	1.10125%	0.96563%	0.77688%	0.71844%	0.70281%	0.72906%
12 month rate	0.99375%	0.84250%	0.83938%	0.91500%	1.04000%	1.20875%	1.17313%	1.02975%	0.83988%	0.77838%	0.76038%	0.79000%

Fig 11: Historical USD LIBORs Table

It consists of 12 rows, with 12 variables each.

Analysis of USD LIBOR

Using Excel's built-in visualisation tools, I created graphs for the LIBOR rates for the different starting dates in *D*, against their months to maturities.

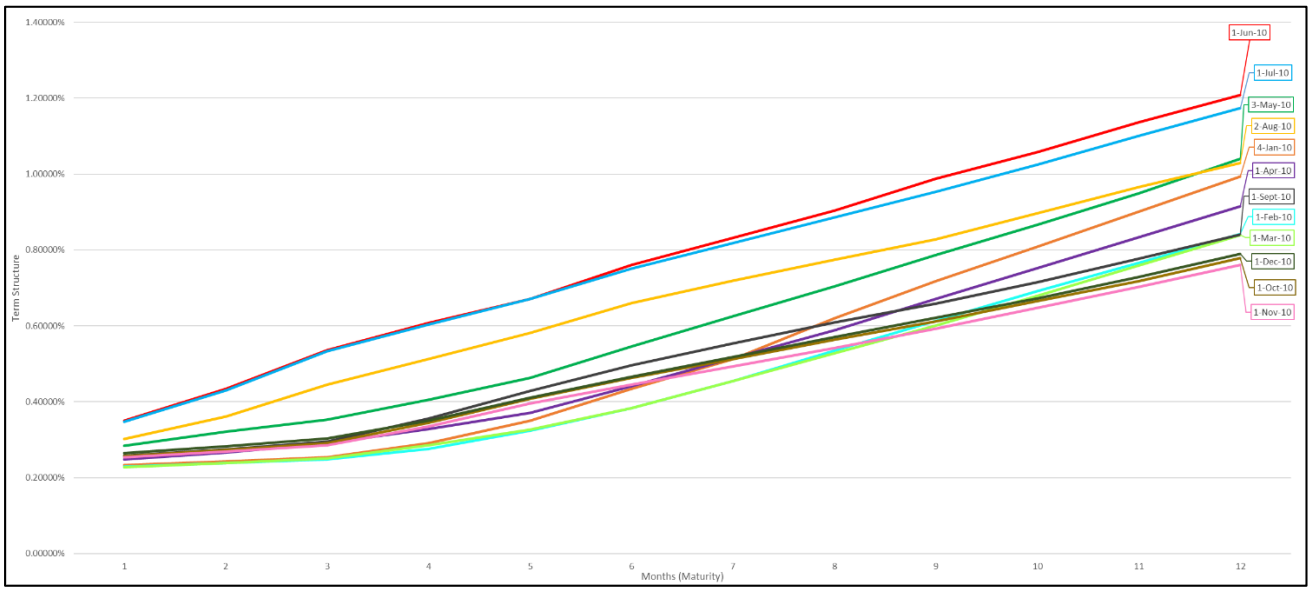

Fig 12: Compiled LIBOR Rates Graphs

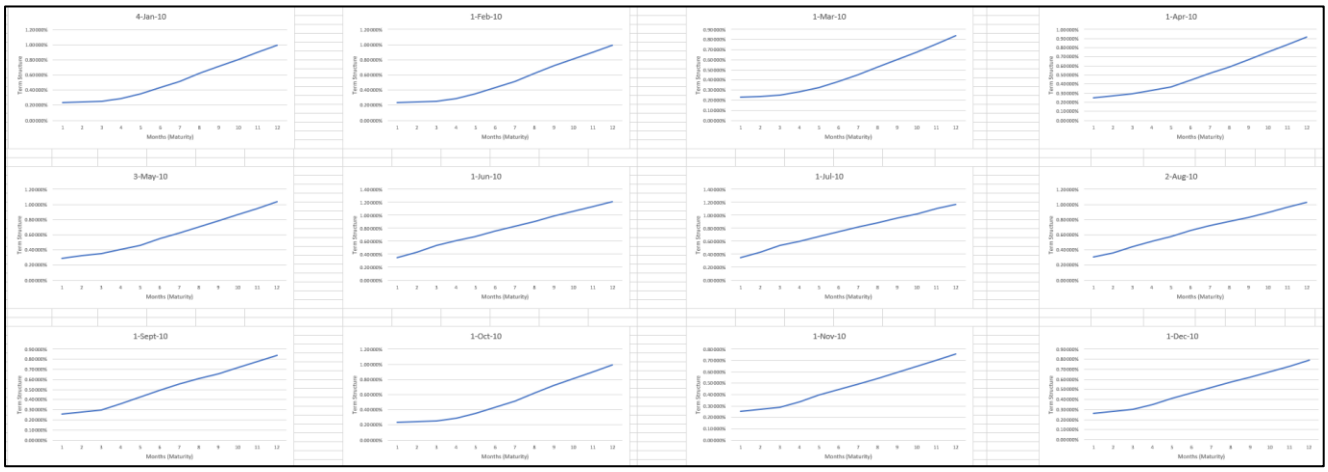

Fig 13: Individual LIBOR Rates Graphs for each starting date in *D*

All 12 term structures exhibit a positive slope, showing that rates are lowest for 1-month maturities and progressively higher for 12-month maturities
The consistent positive slope reflects the market's expectation of economic normalisation and growth throughout 2010. The higher long-term rates reflect a substantial Term Premium, compensating lenders for duration risk and future inflation uncertainty.

The rates for maturities of 1 to 3 months are all low and are clustered together
The short end of the curve is anchored by the U.S. Federal Reserve's Zero Lower Bound (ZLB) policy, which keeps the target Federal Funds Rate near zero. This provides ample liquidity to sustain economic recovery while suppressing the cost of very short-term funding.

The rates for the months of June and July are consistently the highest across all the maturities.
The high rates could possibly be a result of the European Sovereign Debt Crisis that also peaked around this time. This crisis caused global deleveraging and extreme counterparty risk aversion, leading U.S. banks to hoard USD liquidity. This abrupt increase in funding risk translated directly into higher LIBOR rates.

The rates for the first 5 months of the year (Jan, Feb, Mar, Apr, May) and the last 4 months of the year (Sept, Oct, Nov, Dec) have an upward sloping trend where there exists a turning point for which the slope becomes steeper. Generally, this turning point is at the 3 or 4 month maturity.
The presence of the turning point suggests that the market had speculations of rate accelerating sharply. This likely reflected the market pricing a period of near-term stability before rates needed to climb to compensate for longer-term inflation expectations.

Next, we append set *D* with the following dates:

- January 03 2011
- February 01 2011
- March 01 2011
- April 01 2011
- May 02 2011
- June 03 2011
- July 01 2011
- August 01 2011
- September 01 2011
- October 03 2011
- November 01 2011
- December 02 2011

This new set is denoted as Set *E*

By using the new dates in Set E, we can use Built-In excel functions to calculate the number of days for each maturity. For each Starting Date, we obtain the number of days for each maturity period as the variable *DaysInTerm.* For Convenience, I created an excel VBA function to calculate the days in each period.

We append the original data table with the newly obtained *DaysInTerm* to get the following table:

D	4-Jan-10		1-Feb-10		1-Mar-10		1-Apr-10		3-May-10		1-Jun-10	
Maturity(month)	Term Structure	DaysInTerm	Term Structure2	DaysInTerm	Term Structure3	DaysInTerm	Term Structure4	DaysInTerm	Term Structure5	DaysInTerm	Term Structure6	DaysInTerm
1 month rate	0.23344%	28	0.22906%	28	0.22813%	31	0.24863%	32	0.28469%	29	0.35088%	30
2 month rate	0.24344%	56	0.23906%	59	0.23938%	63	0.26694%	61	0.32125%	59	0.43438%	62
3 month rate	0.25438%	87	0.24906%	91	0.25169%	92	0.29150%	91	0.35313%	91	0.53625%	92
4 month rate	0.29125%	119	0.27625%	120	0.28563%	122	0.32850%	123	0.40500%	121	0.60775%	122
5 month rate	0.35025%	148	0.32438%	150	0.32669%	154	0.37063%	153	0.46375%	151	0.67119%	153
6 month rate	0.43438%	178	0.38375%	182	0.38375%	184	0.44156%	183	0.54578%	182	0.76113%	183
7 month rate	0.51250%	210	0.45563%	212	0.45481%	214	0.51563%	214	0.62531%	212	0.83213%	216
8 month rate	0.62000%	240	0.53438%	242	0.52750%	245	0.58906%	244	0.70375%	245	0.90363%	245
9 month rate	0.71813%	270	0.61313%	273	0.60138%	275	0.67125%	277	0.78688%	274	0.98719%	273
10 month rate	0.80875%	301	0.69125%	303	0.67931%	308	0.75294%	306	0.86719%	302	1.05863%	304
11 month rate	0.90063%	331	0.76563%	336	0.75938%	337	0.83375%	334	0.94906%	333	1.13675%	335
12 month rate	0.99375%	364	0.84250%	365	0.83938%	365	0.91500%	365	1.04000%	364	1.20875%	367

1-Jul-10		2-Aug-10		1-Sept-10		1-Oct-10		1-Nov-10		1-Dec-10	
Term Structure7	DaysInTerm	Term Structure8	DaysInTerm	Term Structure9	DaysInTerm	Term Structure10	DaysInTerm	Term Structure11	DaysInTerm	Term Structure12	DaysInTerm
0.34719%	32	0.30281%	30	0.25781%	30	0.25688%	31	0.25375%	30	0.26531%	33
0.43063%	62	0.36094%	60	0.27469%	61	0.27359%	61	0.26859%	63	0.28250%	62
0.53331%	92	0.44469%	91	0.29563%	91	0.29063%	94	0.28594%	92	0.30344%	90
0.60338%	123	0.51213%	121	0.35594%	124	0.34469%	123	0.33500%	120	0.35031%	121
0.67163%	153	0.58125%	154	0.42819%	153	0.40813%	151	0.39625%	151	0.41094%	152
0.75100%	186	0.65994%	183	0.49625%	181	0.46313%	182	0.44563%	182	0.46656%	184
0.81819%	215	0.71944%	211	0.55350%	212	0.51344%	213	0.49350%	214	0.51875%	212
0.88656%	243	0.77375%	242	0.60888%	243	0.56375%	245	0.54125%	242	0.57125%	243
0.95375%	274	0.82781%	273	0.65813%	275	0.61203%	273	0.59313%	273	0.62156%	274
1.02500%	305	0.89656%	305	0.71575%	303	0.66531%	304	0.64750%	304	0.67313%	306
1.10125%	337	0.96563%	333	0.77688%	334	0.71844%	335	0.70281%	336	0.72906%	335
1.17313%	365	1.02975%	364	0.83988%	365	0.77838%	367	0.76038%	365	0.79000%	366

Fig 14: Historical USD LIBORs Table with *DaysInTerm*

Stage 2: Swap Rate of a One-Year Interest Rate Swap

Using the updated LIBOR table (Fig 14.) and an excel VBA user-defined function, I calculated the swap rates (actual/360) of one-year interest rate swap, with swaps happening at monthly intervals.

I made use of the following formulas:

1. **Final Discount Factor, $d_{0,12}$**

$$d_{0,12} = \frac{1}{(1+r_{12}*\frac{n_{12}}{360})}$$

Where:

- n_{12} is the total number of days for the 12 month maturity
- r_{12} 12 month LIBOR from the starting date.

2. **Annuity Factor**

$$\sum_{i=1}^{12}\left(d_{0,i} * \frac{n_i}{360}\right)$$

Where:

- n_i is the total number of days in the ith payment period only

3. **Swap Rate**

$$Swap\ Rate = \frac{1 - Final\ Discount\ Factor}{Annuity\ Factor}$$

Computing the Swap Rates for the dates in *D*:

Swap Rates for One-Year Interest Rate Swaps (%)											
4-Jan-10	1-Feb-10	1-Mar-10	1-Apr-10	3-May-10	1-Jun-10	1-Jul-10	2-Aug-10	1-Sept-10	1-Oct-10	1-Nov-10	1-Dec-10
0.987483081	0.838006479	0.834884174	0.909730935	1.033350479	1.200175131	1.165096343	1.02355558	0.835638705	0.774736057	0.75690247	0.786249916

Fig 15: Table of Swap Rates for dates in *D*

Analysis of Swap Rates

Using Excel's built-in visualisation tools, I created the graph for Swap Rates for One-Year Interest Rate Swaps against their respective Starting Dates.

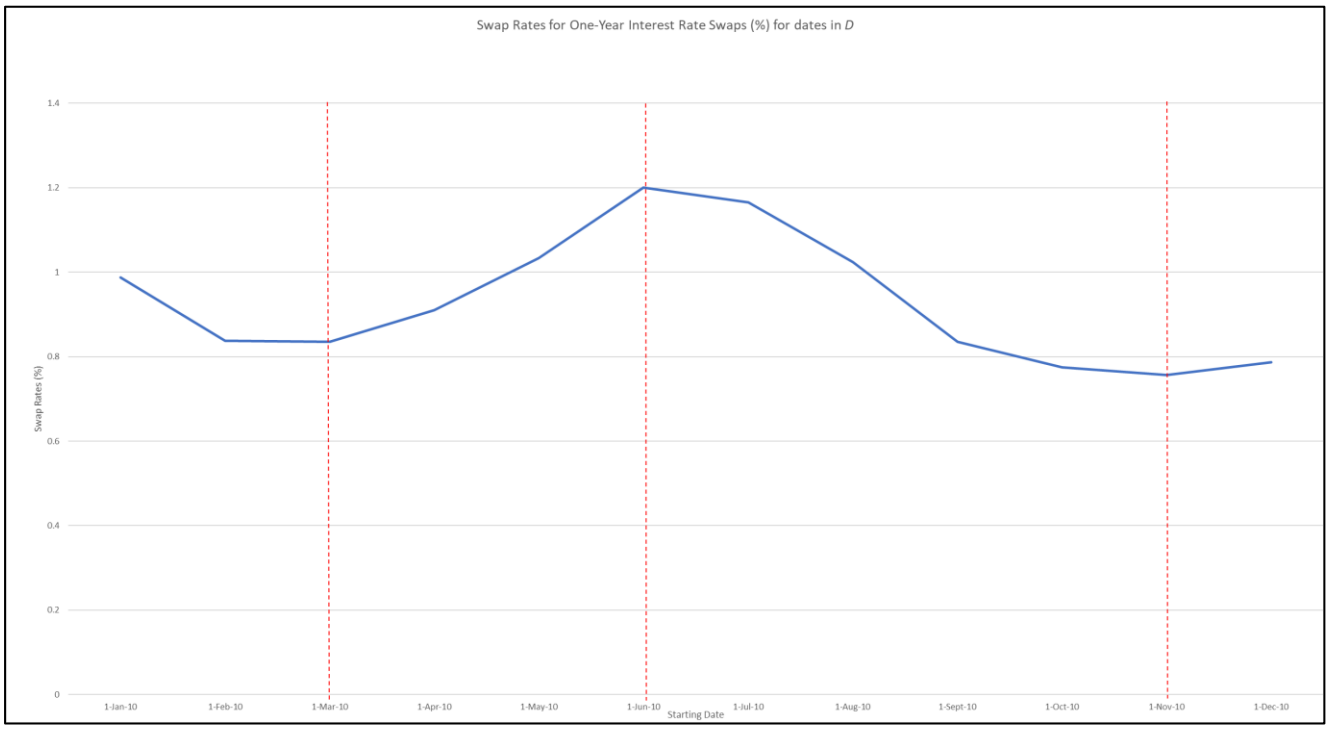

Fig 16: Graph of Swap Rates against Starting Dates

Reference Months	Analysis
January to March	From January to March, there was a decreasing trend in the swap rates. There is a significant decrease in swap rates from January to February, while the swap rates for February and March were almost the same This period saw the initial severe escalation of the European Sovereign Debt Crisis. Investors, fearing global financial contagion, started selling riskier assets and buying 'safer' U.S. Treasury securities. The resulting surge in demand for Treasuries drove their yields down. Since swap rates are priced off Treasury yields, the LIBOR term structure and the 1-year swap rate fell sharply. The stabilisation in March indicates a temporary cooling of the crisis fears as the immediate panic subsided.
March to June	From March to June, there was an increasing trend in the swap rates, with swap rates hitting its highest rate in June This increase was fuelled by strong Q1 corporate earnings and improving U.S. economic data. This boosted the market's optimism for a sustained recovery. This led to speculation that the Federal Reserve would soon begin exiting its zero-rate policy and begin raising the Fed Funds Rate. Swap rates rose sharply to price in an earlier and steeper tightening cycle to combat expected future inflation resulting from the prior QE liquidity injections.
June to November	From June to November, rates fell again, with the steepest change in rates being from July to September. After which, the decrease in rates began to slow down. swap rates in November were the lowest for the entire year. Economic growth slowed sharply in this period, leading to fears of another recession. Investors, seeking security amid heightened risk, shifted capital into U.S. Treasuries, driving down risk-free yields and further pressuring swap rates downwards. The slowing economic growth and fears also cause the market expectations for rate hikes to vanished entirely. Instead, attention turned to the Federal Bank initiating a second round of Quantitative Easing (a second round of bond purchases). This anticipation forced long-term interest rate expectations—and the 1-year swap rate—to their lowest point, culminating in November when the Federal Bank formally announced the QE2 program.
November to December	From November to December there was a slight increase in swap rates The confirmation of QE2 signalled the Fed's commitment to maintaining liquidity and removed the immediate uncertainty in the markets. This stabilisation led to a slight return of risk appetite and improved liquidity in the market. Short-term rates rose slightly higher off their historical lows as the market prepared for year-end and modestly improved its outlook for 2011.

Overall, Swap Rates in 2010 were consistently lower than current rates in 2025

The highest swap rate observed in 2010 was for June 2010, where it was approximately 1.2%. By comparison, similar maturity USD swap rates in the current market (2025) are significantly higher, often exceeding 3% to 4%. This shows that the low interest rates were a direct effect of the Federal Bank's QE measures and zero-rate policy, and that these measures were effective in suppressing interest rates in the market.

The difference in the rates also reflects the difference in market environments. From one focused on stimulating demand and preventing deflation to one that is focused on combating inflation.

Stage 3: Tracking Swap Value of a One-Year Interest Rate Swap

In this section, we track the swap value on the first business day on the following 11 months for the one-year interest rate swap beginning on **January 04 2010**. This value is computed for 1 USD notational principal for the party paying floating payments and receiving fixed payments.

Using the updated LIBOR table (Fig 14.) and an excel VBA user-defined function, I calculated the swap value (actual/360) of the one-year interest rate swap.

I made use of the following formulas:

1. **Present Value of Fixed Leg**

$$Swap\ Rate * \sum_{i=1}^{12-k}\left(DF(i) * \frac{n_i}{360}\right)$$

Where:
- DF(i) is the discount factor for the ith remaining payment date
- n_i is the total number of days in the ith remaining payment period
- K is the number of months that have elapsed since the swap's inception.

2. **Present Value of the Floating Leg**

$$1 - DF(T_n)$$

Where:
- DF(T_n) is the discount factor for the final maturity date

3. **Swap Value**

$$V = PV(Fixed) - PV(Floating)$$

Computing the swap values for January 04 2010 gives:

Swap Value for Swap on 4-Jan-10 ($)	
K	Swap Value
1	0.0020965
2	0.002651243
3	0.002443974
4	0.001940002
5	0.000943121
6	0.001227236
7	0.001738626
8	0.002174521
9	0.001818881
10	0.001257684
11	0.000661831

Fig 17: Table of Swap Values

Analysis of Swap Value

Using Excel's built-in visualisation tools, I created the graph for the swap value for the one-year interest rate, against varying values of *K*.

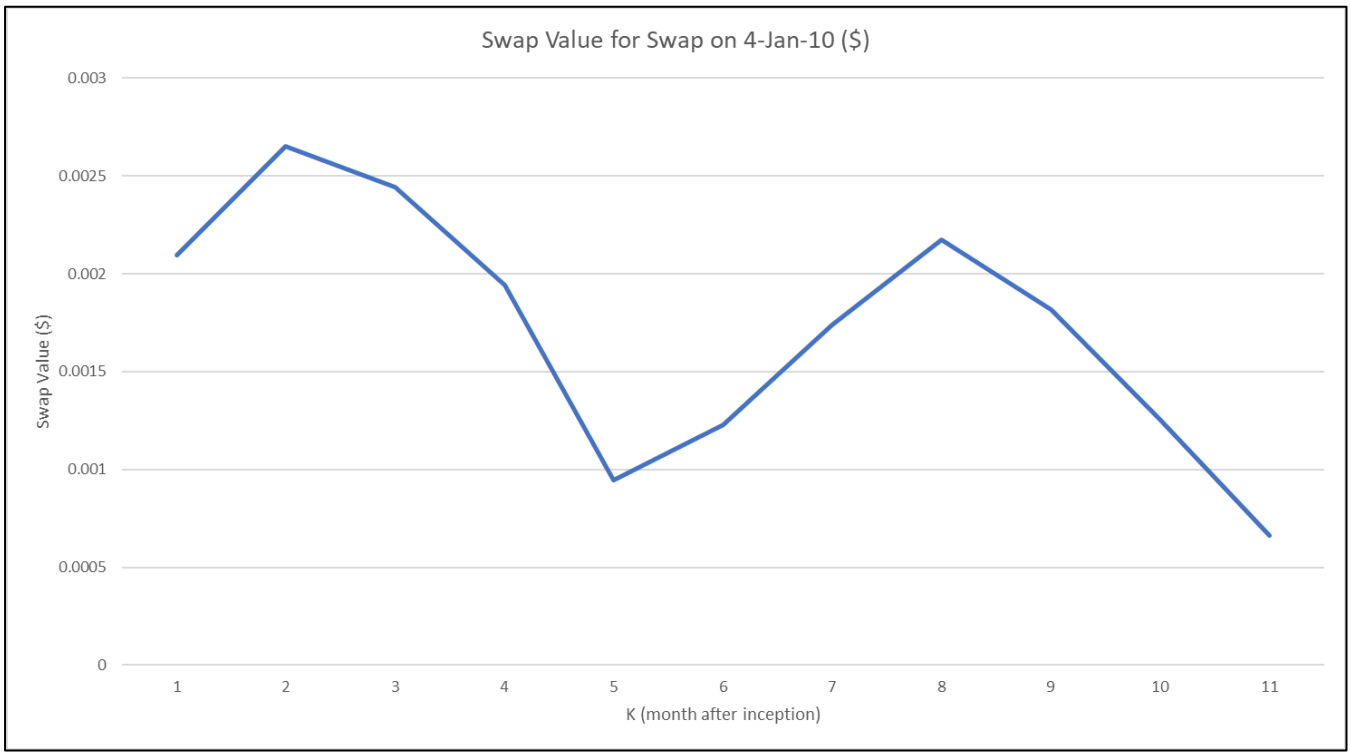

Fig 18: Graph of Swap Value against Varying *K*

Values of K	Analysis
K = 1 to K = 2	From K = 1 to K = 2, the swap value has an increasing trend. The swap value at K=2 is the highest that it will achieve. This suggests the prevailing short-term and mid-term LIBOR rates were likely declining in the first two months of 2010. This is consistent with the effects of European Sovereign Debt Crisis on the market, where there was a sharp fall in the LIBOR rates in January and February. This led to the observed effect of rising swap value.
K = 2	There is a turning point at K = 2, marking the end of the initial increasing trend The presence of a turning point at the peak of the swap value indicates that following this, the market expected LIBOR rates to increase. This is consistent with what was happening at this time (March 2010) where there was stabilisation in the market due to the temporary cooling of the crisis fears. This signalled a shift in market sentiment which is reflected in the graph for swap value.
K = 2 to K = 5	From K = 2 to K = 5, the swap value has a decreasing trend A decrease in swap value would suggest that the prevailing LIBOR rates in this period (March - June) were likely rising. This is consistent with the economic situation at that time whereby strong Q1 corporate earnings and improving U.S. economic data lead to a boost in market optimism. Furthermore, market speculation led to swap rates rising sharply to price in an earlier and steeper tightening cycle to combat expected future inflation resulting from the prior QE liquidity injections. The low swap value is consistent with the LIBOR rates in 2010 reaching its highest in June.
K = 5	At K = 5 there is another turning point The presence of a turning point at one of the troughs of the graph indicates that following this, the market expected LIBOR rates to decrease. This is consistent with what was happening at this time (June 2010) where economic growth had slowed sharply and there were rising fears of another recession
K = 5 to K = 8	From K = 5 to K = 8, the swap value has an increasing trend again. An increase in swap value would suggest that the prevailing LIBOR rates in this period (June - September) were likely falling. This is consistent with the economic situation at the time whereby the slowing economic growth coupled with fears of a second recession led to investors shifting capital into U.S Treasuries. Furthermore, expectations of a rate hike also vanished completely. This caused the market swap rates to fall, leading to the observed effect of rising swap value.
K = 8	At K = 8, there is another turning point Again, the presence of the turning point suggests that following this period, the market expected LIBOR rates to increase. This was consistent with what was happening at the time (September) where the market speculated for a second Quantitative Easing Program to be executed by the Federal Bank. Despite the announcement only being formalised in two months later, the market was already beginning to stabilise, signalling a shift in market sentiment and thus a shift in swap rates as well. This led to the observable turning point in the graph for swap value.
K = 8 to K = 11	From K = 8 to K = 11, the swap value has a decreasing trend. The swap value at K = 11 is the lowest it will achieve The decrease in swap value suggests that the prevailing LIBOR rates in this period (September - December) were increasing. Again, this was consistent with what was happening at the time where the confirmation of QE2 lead to stabilisation of the market and an increase in investor's risk appetites. The improved liquidity in the market as well as improved market confidence caused swap rates to rise, leading to the observed decrease in swap value.

Overall, the swap value of this Interest Rate swap is always positive.
The original fixed Swap Rate agreed upon at the start of the swap (January 04, 2010) was consistently higher than the prevailing 1-year market swap rate for the remaining term throughout the entire year of 2010. This is a direct consequence of the low-rate environment established by the Federal Reserve's zero-rate policy and Quantitative Easing (QE) measures post-European Sovereign Debt Crisis. While the original Swap Rate was likely determined by the rate environment just before or at the inception of the swap, the subsequent market rates fell rapidly (especially due to the QE2 anticipation and recession fears). As a result, the original fixed rate would be consistently higher than the floating market rate.

The trends observed for the Swap Value mirrors the trends observed for the Swap Rates for One-Year Interest Rate Swaps in 2010. However, the curve in the Swap Value graph is much steeper than that of the Swap Rate graph.

This is to be expected as the Swap Value for the party receiving fixed rate payments and paying floating rate payments is calculated as the difference in the fixed-leg and the floating-leg. When the market swap rates fall, the swap value would naturally rise. Even though we are using the term structures at each month, it will still be reflective of the market climate at that point in time. As the swap value is calculated by the summation of discount factors applied to a stream of different cash flows, the changes in swap rate would lead to a larger magnitude change in swap value, making its graphical representation noticeably steeper than the underlying rate curve.

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- End of Part 2 -